



GENERATIVE AI

IBM

DAY 1

Session Agenda

Generative AI
Foundations



1

Evolution of AI

From the Turing Test to the GenAI boom

2

Deep Learning & Architectures

CNNs, RNNs and Transformers explained

3

Large Language Models

How LLMs work and what they deliver

4

Generative AI Model Types

VAE · GAN · Transformer · Diffusion

5

Foundation Models

The new paradigm train once, use everywhere

6

Prompt Engineering

Zero-shot, few-shot, chain-of-thought techniques

7

Hands-on Lab

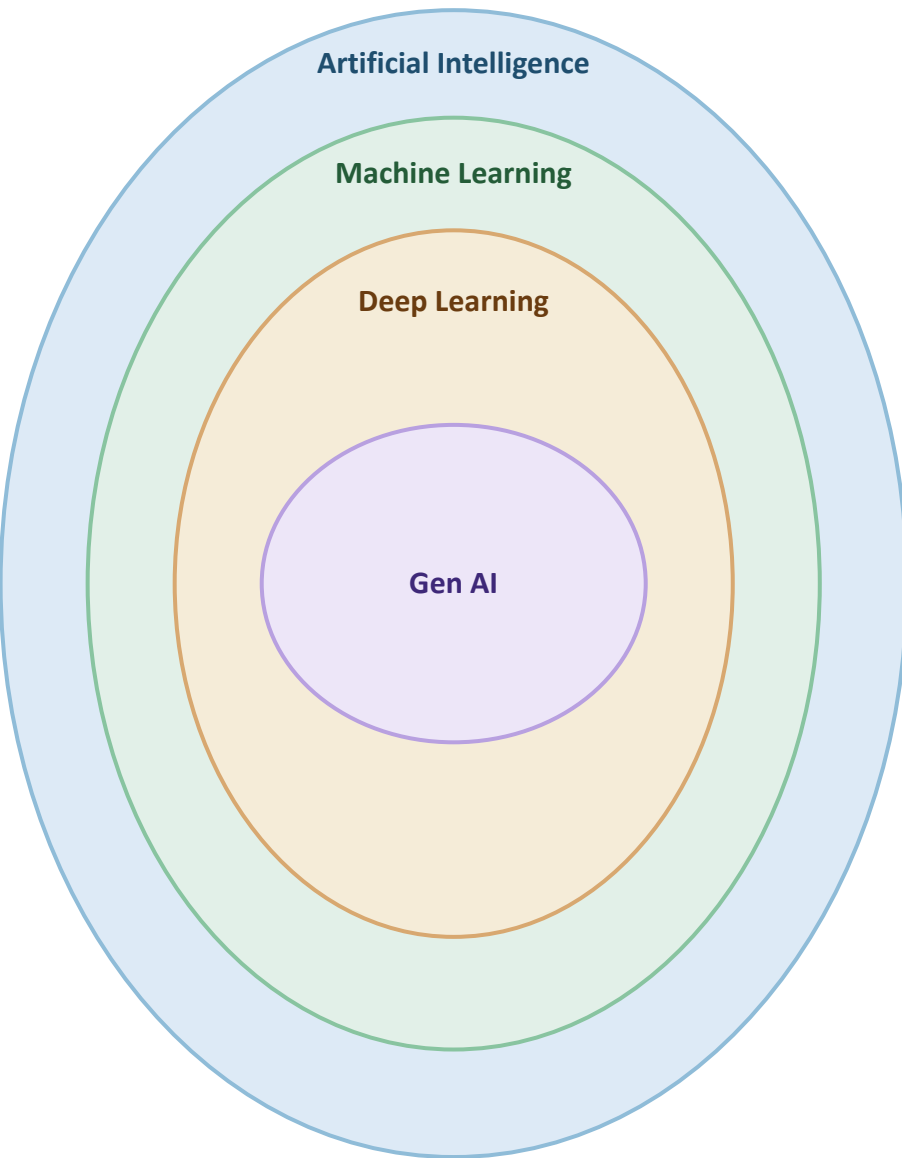
Sentiment · Prompting · Code gen · RAG ·
Language tutor

8

IBM watsonx & Granite

Enterprise AI platform and foundation models

AI Ecosystem: AI → ML → DL → Gen AI



Artificial Intelligence (AI)

Broadest umbrella term

The science of creating systems that perform tasks requiring human intelligence — perception, reasoning, learning, and decision-making.

Examples

Rule-based systems · Chess engines · Recommendation engines · Speech recognition

Machine Learning (ML)

AI that learns from data

A subset of AI where models improve by learning statistical patterns from data, without being explicitly programmed for each task.

Examples

Spam filters · Fraud detection · Price prediction · Supervised / Unsupervised / Reinforcement

Deep Learning (DL)

ML with layered neural networks

Uses Artificial Neural Networks (ANNs) with many hidden layers to learn hierarchical representations. Requires large datasets and GPUs.

Examples

CNNs (computer vision) · RNNs/LSTMs (speech) · Transformers (language, NLP)

Generative AI (GenAI)

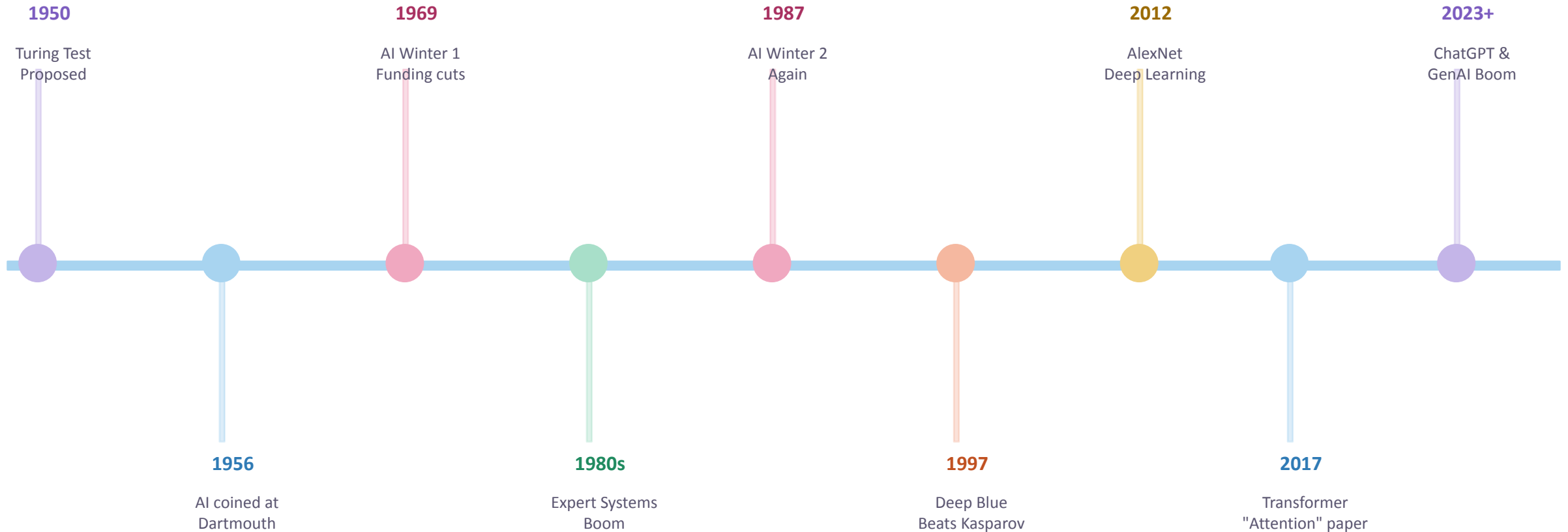
DL that creates new content

Models trained to generate new data resembling their training distribution — text, images, code, audio, and video at enterprise scale.

Examples

ChatGPT · DALL-E · Stable Diffusion · GitHub Copilot · IBM Granite / watsonx.ai

The Evolution of AI



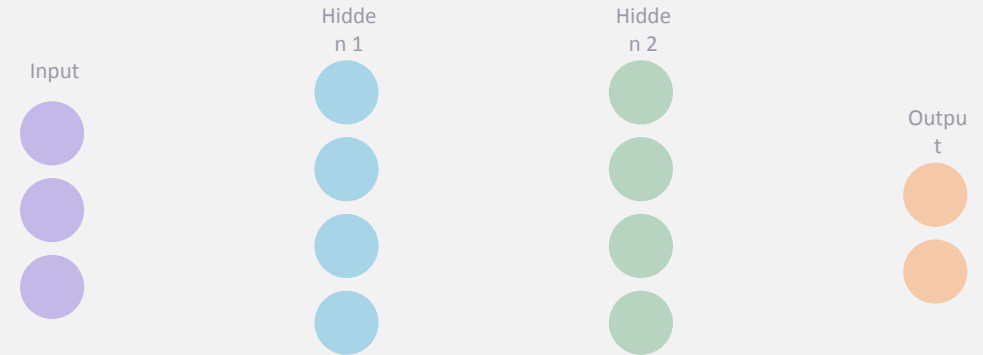
❄️ **AI Winters:** Periods (1969–1980, 1987–1993) where funding and interest collapsed because AI couldn't deliver on over-promised expectations. Each winter was followed by a breakthrough that reignited the field.

What is Deep Learning?

The engine behind every generative AI system

Deep Learning

A branch of AI that uses multi-layered artificial neural networks (ANNs) to learn patterns from large datasets — mimicking how the human brain processes information.



Neurons & Layers

Input → Hidden → Output layers; weights & biases adjust during training.

Training Modes

Supervised (labeled data) for classification; Unsupervised for pattern discovery.

Why It Matters

More layers = more complex patterns. Basis for every Gen AI model today.

Deep Learning Architectures

Three foundational designs - each built for different data types

CNN

Convolutional
Neural Networks

Best for image tasks

Scans images layer by layer detecting shapes and patterns. Used in facial recognition, image classification, and video analysis.

Examples

Face ID · Medical imaging · Autonomous vehicles

RNN

Recurrent
Neural Networks

Best for sequential data

Has memory — remembers previous inputs in a sequence. Designed for speech, text and time-series data.

Examples

Speech-to-text · Translation · Sentiment analysis

Transformer

Transformer-
based Models

Powers all modern LLMs

Uses self-attention to focus on important parts of input — no sequential constraint. Excels at understanding long-range context.

Examples

GPT-4 · IBM Granite · Gemini · Claude

IBM Granite and watsonx.ai are built on Transformer architecture - the most powerful and scalable design for enterprise Gen AI.

Large Language Models (LLMs)

Transformer-based models that perform human-like language tasks at scale

LLMs are trained on massive text datasets (billions to trillions of tokens) and can write, code, translate, summarize, classify, and reason automating knowledge work across your client's organisation.

170T+

Parameters — GPT-4

540B

Parameters — PaLM 2

13B

Parameters — IBM Granite

1T+

Training tokens — Granite

What LLMs deliver for your clients:

1

Content generation

Emails, reports, marketing copy — at scale

2

Code generation

Accelerate developer cycles, explain code

3

Document summarisation

Contracts, call logs, research papers

4

Intelligent Q&A

Customer support, enterprise search

5

Translation

50+ languages — global reach

6

Classification & Sentiment

CRM tickets, customer feedback, fraud detection

4 Core Generative AI Model Types

Each uses distinct architecture to learn from data and create new content

VAE

Variational Autoencoders

How it works:

Encoder compresses input → latent space → Decoder reconstructs and generates new outputs

Business use:

Image synthesis, anomaly detection, drug discovery, financial risk modelling

GAN

Generative Adversarial Networks

How it works:

Generator creates fakes → Discriminator detects fakes → Both improve through competition

Business use:

Marketing creative assets, synthetic data, deepfake detection, geospatial generation

Transformer

Transformer-based Models

How it works:

Self-attention mechanism finds relationships across entire input — no sequential constraint

Business use:

Backbone of GPT-4, IBM Granite, Gemini, Claude — language, code and enterprise AI

Diffusion

Diffusion Models

How it works:

Forward: add noise to data → Reverse: iteratively denoise to generate clean output

Business use:

DALL-E 2, Stable Diffusion XL, Imagen — creative image and video generation

Transformers: the game changer

Every model you've heard of - GPT, BERT, LLaMA, Gemini, Granite is a Transformer

Self-attention

Every word computes relevance to every other word simultaneously. In “The cat sat because it was tired” - model knows “it” = “cat” instantly.

Positional encoding

Adds unique numerical patterns to each token to preserve word order, since all tokens are processed in parallel.

Parallel processing

Entire sequence processed at once on GPUs. Orders of magnitude faster than sequential RNNs. Enables training on trillions of tokens.

Encoder vs Decoder

Encoder-only (BERT): understands text. Decoder-only (GPT, LLaMA, Granite): generates text. Encoder-decoder (T5): both.

“Transformers read the entire page at once and highlight connections. That’s why ChatGPT can write coherent essays.”

Foundation Models - The New Paradigm

Trained once on massive data, adapted for unlimited tasks

Foundation Model

A large general-purpose AI model pre-trained on vast datasets - text, images, code - using self-supervised learning. Adapted via prompting or fine-tuning for many tasks without retraining.

Old approach:

Build a NEW model from scratch for each task - costly, slow, requires labeled data.

New approach:

ONE foundation model → prompt or fine-tune → unlimited use cases.

Three defining properties:

Pre-trained at scale

Trained on trillions of tokens - text, images, code - without task-specific labels. Broad world knowledge from day one.

Multimodal capability

A single model can understand and generate text, images, audio, video, and code - flexible enough for any client use case.

Adaptable & reusable

Deploy for hundreds of tasks via prompting, fine-tuning, or RAG - without expensive retraining for every new application.

Foundation Model Landscape

Key models your clients are asking about - and where IBM fits

Model	Vendor	Params	Architecture	Key Use Cases
GPT-4 / GPT-4o	OpenAI / Microsoft	~1T+	Transformer	General language, reasoning, coding, vision
Gemini / PaLM 2	Google DeepMind	540B+	Transformer	Multilingual, multimodal, reasoning, science
IBM Granite (13B)	IBM Research	13B	Decoder-only	NLP, code, enterprise tasks - governance-ready
LLaMA 2 / LLaMA 3	Meta AI	7B – 70B	Transformer	Open-source, fine-tuning, commercial research
Falcon-7B-Instruct	TII	7B	Decoder-only	English text gen, chatbots, educational tools
DALL·E / Stable Diffusion	OpenAI / Stability	N/A	Diffusion	Text-to-image generation, creative workflows
IBM Granite is the only foundation model family built specifically for enterprise governance, auditability, and compliance - available on watsonx.ai.				

Prompt engineering

Same model, dramatically different results — the difference is how you ask

✗ Bad prompt

"Tell me about AI"

- Generic, rambling, unfocused
- No structure, no constraints

✓ Good prompt

"Act as a professor. Explain CNN, RNN, Transformer to an undergrad. One example each. Under 200 words."

- Focused, structured, useful

Key techniques for your classroom

Zero-shot Ask directly, no examples needed

Few-shot Provide 2-3 examples first

Role setting Prime the model with expertise

Chain-of-thought Force step-by-step reasoning

Output format Specify structure explicitly

"Classify this review as positive or negative"

"Positive: Loved it. Negative: Terrible. Now classify: It was okay"

"Act as a senior Python developer"

"Think step by step before answering"

"Respond in JSON with fields: sentiment, confidence"



Hands-on Lab - Foundation Models

1

Sentiment detection

zero-shot classifier - one instruction, instant classifier

2

Good vs bad prompts

Same question, dramatically different results - the "aha" moment

3

Code generation

Python Fibonacci with Granite-34b - enterprise code gen

4

Grounded Q&A (RAG)

Model answers ONLY from provided text foundation of enterprise AI

5

Language tutor

One prompt creates vocab builder + translator + quiz generator

IBM watsonx - The Enterprise AI Platform

Three integrated products for building, governing, and scaling Gen AI

watsonx.ai

Build & deploy

Platform for foundation models, ML, and generative AI. Access IBM Granite models, open-source models from Hugging Face, and a Prompt Lab for rapid experimentation.

Key capabilities:

- Prompt engineering & tuning
- Model training & fine-tuning
- AI application development

watsonx.data

Data & retrieval

Open, hybrid, governed data store built for AI workloads. Powers RAG (Retrieval-Augmented Generation) so models can access live, current enterprise knowledge.

Key capabilities:

- Connect AI to live enterprise data
- Reduce hallucinations via RAG
- Unified data governance

watsonx.governance

Trust & compliance

Monitor AI for bias, drift, and fairness. Full model lineage, auditability, and regulatory compliance — critical for banking, healthcare, and public sector clients.

Key capabilities:

- Bias & fairness monitoring
- Explainability & audit trails
- Regulatory compliance (EU AI Act, GDPR)

IBM Granite Foundation Models

Enterprise-grade - built for business reliability and trust

granite.13b.base

Base model

Trained on 1T+ tokens. Decoder-only architecture. Foundation for all variants.

granite-13b-instruct

Instruct variant

Fine-tuned with SFT on Flan, Dolly, Anthropic guidelines. Best for task instructions.

granite-13b-chat

Chat variant

Refined with Contrastive Fine-Tuning (CFT). Optimised for dialogue and conversation.

Why IBM Granite for Enterprise?

E

Efficient at scale

13B params runs on a single V100-32GB GPU — cost-effective to deploy.

G

Governance built-in

HAP content filter, full lineage & auditability via watsonx.governance.

V

Versatile for business

Summarisation, classification, Q&A, code gen — proven on enterprise tasks.

T

Transparent training

IBM publishes training data details — unlike proprietary black-box models.

Hugging Face the GitHub of AI

Where your students will go to experiment

250K+

Open models

50K+

Datasets

1M+

Demos

50K+

Companies

Model Hub

Search & deploy pre-trained models for text, image, audio, video

Transformers Library

Compatible with PyTorch, TensorFlow, and JAX — the industry standard

Spaces

Host interactive AI demos — great for student projects

HuggingChat

Open-source ChatGPT alternative your students can explore freely

IBM Partnership

watsonx.ai uses HF models; HF hosts IBM open-source LLMs

SOC 2 compliant • Free tier available • Academic-friendly licensing

Key takeaways

- ✔ Transformers (2017) are the foundation of ALL modern GenAI
- ✔ Four model types - VAE, GAN, Transformer, Diffusion - each has a sweet spot
- ✔ Foundation models: train once, adapt to any task with minimal fine-tuning
- ✔ Prompt engineering is the most immediately useful GenAI skill
- ✔ IBM watsonx: build (ai), feed (data), govern (governance)
- ✔ Hugging Face: 250K+ models - where students should experiment
- ✔ Always verify GenAI output - hallucination is real
- ✔ GenAI doesn't replace faculty - it empowers them



Questions & discussion

*“GenAI isn’t something happening to education —
it’s something happening in education.”*